Software Requirements Specification (SRS)

Arati Business ERP

Document Version: 2.0
Status: Software Requirements Baseline — Draft for Implementation
Project: Arati Business ERP
Primary Scope: Distribution + Wholesale + Retail/POS + Inventory ERP
Accounting Integration: TallyPrime
Priority: ERP first; public website later

---

1. Purpose

This Software Requirements Specification defines the software behavior, functional requirements, non-functional requirements, acceptance expectations, and system constraints for the Arati Business ERP.

This version supersedes the earlier SRS v1.0 baseline and is aligned with:

- `BRD.md`
- `BUSINESS-RULES_v1.0.md`
- `ARCHITECTURE_v2.0.md`
- `DATABASE_v1.0.md`
- `TECHSTACK_v1.1.md`
- `PROTECTION.md`

The SRS is the bridge between business needs and implementation.

It should be specific enough that each important requirement can become:

Requirement
↓
Use Case
↓
API / UI behavior
↓
Implementation
↓
Automated Test

↓
Acceptance

The ERP must remain generic and configurable. It must not hard-code behavior around any particular brand, supplier, customer, dealer, product, or model.

---

2. Product Vision

The ERP shall provide one operational system for:

Procurement
↓
Goods Receipt
↓
Inventory
↓
Retail / Wholesale / Dealer Sales
↓
Payments / Receivables / Payables
↓
Returns / Fulfilment
↓
Delivery / Installation / Warranty
↓
Reports / Audit
↓
TallyPrime Integration

The ERP is the operational source of truth.

TallyPrime remains the accounting platform for the initial release.

The public website is a later product boundary and must not complicate the ERP's core transaction model.

---

3. Scope

3.1 In Scope

Identity and access

- Authentication
- Session management
- User management
- Roles
- Permissions
- Organization/branch/warehouse access scope
- Security audit

Master data

- Organizations
- Branches
- Warehouses
- Warehouse locations
- Brands
- Categories
- Units
- Tax configuration
- Products
- Product variants
- Parties
- Party roles
- Payment terms
- Price lists

Procurement

- Purchase orders
- Goods receipts
- Purchase invoices
- Partial receiving
- Batch capture
- Serial capture
- Procurement traceability

Inventory

- Stock ledger
- Stock balances
- Serial inventory
- Batch inventory
- Stock transfers
- Stock adjustments
- Stock availability
- Inventory valuation foundation
- Stock history

Sales

- Retail/POS sales
- Wholesale sales
- Dealer sales
- Quotations
- Sales orders
- Sales invoices
- Discounts
- Pricing
- Credit sales

Financial operations

- Payments
- Receipts
- Advances
- Payment allocation
- Receivables
- Payables
- Payment history

Returns

- Sales returns
- Purchase returns
- Return eligibility
- Return disposition
- Serial/batch traceability

Fulfilment

- Delivery
- Partial fulfilment foundation
- Installation foundation
- Warranty foundation

Control and integration

- Audit logs
- Notifications
- Attachments
- Integration outbox
- TallyPrime synchronization
- Tally reconciliation foundation
- Operational reports

---

4. Out of Scope for Initial MVP

The following are explicitly not part of the initial ERP scope unless separately approved:

- Full accounting replacement for TallyPrime
- Payroll/HR
- Advanced CRM
- AI forecasting
- Native mobile application
- Public e-commerce
- Marketplace integrations
- Advanced service management
- Marketing automation
- Complex multi-company accounting
- Microservice decomposition
- Offline POS synchronization
- Dedicated search infrastructure such as Elasticsearch/OpenSearch
- Complex workflow engines
- Full manufacturing/MRP
- Fleet management
- Advanced field-service management

These can be future extensions without changing the core data model where practical.

---

5. User Roles

The system shall use configurable permissions rather than permanently hard-coded role capabilities.

Suggested initial roles:

Role	Typical responsibility
Owner	Full business oversight
Administrator	System administration
Manager	Operations and management
Sales Staff	Retail/wholesale/dealer sales
Inventory Manager	Stock operations
Accountant	Payments and financial coordination

Delivery Staff	Delivery/fulfilment

Actual role names and authority remain configurable.

---

6. Authorization Model

FR-AUTHZ-001 — Authentication

Protected ERP functions shall require authentication.

FR-AUTHZ-002 — Permission Enforcement

Every protected operation shall be authorized by the backend.

FR-AUTHZ-003 — Object-Level Authorization

The backend shall verify that the authenticated user is allowed to access the specific organization, branch, warehouse, or other scoped record.

FR-AUTHZ-004 — No Frontend Trust

Hiding a button or page shall never be considered authorization.

FR-AUTHZ-005 — Sensitive Operations

The following shall require elevated permissions and may require approval:

- Stock adjustment
- Transaction cancellation
- Payment reversal
- Price override
- Large discount
- User management
- Permission management
- Tally configuration
- Tally reconciliation/retry
- Important master-data changes

Exact approval thresholds are **TBD — Business Confirmation Required**.

---

7. Authentication Requirements

FR-AUTH-001 — Login

Registered users shall be able to securely authenticate.

FR-AUTH-002 — Logout

Users shall be able to terminate their session.

FR-AUTH-003 — Password Change

Users shall be able to change their password through the supported flow.

FR-AUTH-004 — Password Recovery

The system shall support secure password recovery where enabled.

FR-AUTH-005 — Account Status

Authorized administrators shall be able to activate/deactivate user accounts.

FR-AUTH-006 — Session Security

Sessions shall expire and be invalidated according to the security policy.

FR-AUTH-007 — Security Audit

Important authentication/security events shall be auditable.

---

8. Organization, Branch and Warehouse

FR-ORG-001 — Organization

The system shall maintain one or more organizations according to deployment requirements.

FR-ORG-002 — Branches

Authorized users shall be able to create, update, activate, and deactivate branches.

FR-ORG-003 — Warehouses

Authorized users shall be able to manage warehouses under branches.

FR-ORG-004 — Locations

The system shall support optional physical locations inside warehouses.

FR-ORG-005 — Inventory Ownership

Stock shall belong to an organization and warehouse/location scope.

FR-ORG-006 — Scope Enforcement

Users shall only access locations they are authorized to access.

FR-ORG-007 — Future Expansion

Adding branches or warehouses shall not require redesign of the transaction model.

FR-ORG-008 — Explicit Transfers

Cross-warehouse or cross-branch stock movement shall be represented as an explicit transfer.

---

9. Master Data Requirements

9.1 Brands

FR-BRAND-001

Authorized users shall be able to create, edit, activate, and deactivate brands.

FR-BRAND-002

Brand records shall contain data, not brand-specific business logic.

9.2 Categories

FR-CAT-001

Authorized users shall be able to create and manage categories.

FR-CAT-002

The system shall support hierarchical categories/subcategories.

FR-CAT-003

Inactive categories shall not normally be selectable for new transactions.

9.3 Units

FR-UNIT-001

The system shall support configurable units such as:

- Piece
- Box
- Set
- Meter
- Other approved units

FR-UNIT-002

Units shall define supported quantity precision.

9.4 Tax Configuration

FR-TAX-001

Authorized users shall be able to maintain applicable tax configuration.

FR-TAX-002

Products/variants shall be associable with applicable tax rules.

FR-TAX-003

Tax calculations shall use the approved tax configuration.

FR-TAX-004

The software shall not assume a final GST structure until business requirements are confirmed.

Status: GST/tax model is **TBD — Business Confirmation Required**.

---

10. Product Management

FR-PROD-001 — Product Creation

Authorized users shall be able to create products.

FR-PROD-002 — Product Identity

Each stockable/sellable variant shall have a unique SKU within the approved organization scope.

FR-PROD-003 — Product Fields

Where applicable, products/variants shall support:

- Name
- SKU
- Brand
- Category
- Model/variant
- Barcode
- HSN
- Tax rule
- Unit
- Purchase cost
- MRP
- Reorder level
- Description
- Attributes
- Product image/attachment

FR-PROD-004 — Tracking Type

The system shall support:

QUANTITY
SERIAL
BATCH
HYBRID

FR-PROD-005 — Product Status

Products/variants shall support controlled active/inactive status.

FR-PROD-006 — Historical Integrity

Deactivating or changing a product shall not invalidate historical transactions.

FR-PROD-007 — Search

Users shall be able to search by relevant identifiers including:

- Name
- SKU
- Barcode
- Model
- Serial number where applicable

---

11. Product Variants

FR-VAR-001

A product may contain multiple variants/models.

FR-VAR-002

A variant shall have independent identity where inventory, SKU, barcode, pricing, tax, or tracking differs.

FR-VAR-003

Historical transaction lines shall retain the price and commercial values actually used at posting time.

Changing current price configuration shall not rewrite historical transactions.

---

12. Party Management

FR-PARTY-001 — Central Party Model

The ERP shall maintain customers, dealers, suppliers, distributors, and other business relationships using one party model.

FR-PARTY-002 — Multiple Roles

One party may have multiple roles.

Candidate roles:

CUSTOMER
DEALER
SUPPLIER
DISTRIBUTOR
OTHER

FR-PARTY-003 — Party Data

Where applicable, parties shall support:

- Name
- Display name
- Phone
- Email
- Address
- GSTIN
- Credit limit
- Payment terms
- Status
- Transaction history

FR-PARTY-004 — Party History

Users shall be able to view authorized purchase, sales, payment, return, and outstanding history for a party.

FR-PARTY-005 — Outstanding

The system shall calculate or expose party outstanding balances from authoritative transaction/allocation data.

---

13. Pricing

FR-PRICE-001 — Multiple Price Lists

The system shall support multiple price lists.

Examples:

RETAIL
WHOLESALE
DEALER
PROMOTIONAL
SPECIAL PARTY

FR-PRICE-002

A product variant may have different prices in different price lists.

FR-PRICE-003

Prices shall support effective dates where required.

FR-PRICE-004

Party-specific price lists shall be supported.

FR-PRICE-005

Quantity-based pricing shall be supported where required.

FR-PRICE-006

Discounts shall not overwrite base price configuration.

FR-PRICE-007

The system shall record the final unit price/discount actually used on the posted transaction.

FR-PRICE-008

Pricing precedence shall be deterministic.

Exact precedence is **TBD — Business Confirmation Required**.

---

14. Procurement

14.1 Purchase Orders

FR-PUR-001

Authorized users shall be able to create purchase orders.

FR-PUR-002

A purchase order shall support:

- Supplier
- Warehouse
- Products
- Quantities
- Prices
- Discounts
- Tax
- Notes
- Status

FR-PUR-003

Purchase orders shall support controlled lifecycle states.

Candidate states:

DRAFT
CONFIRMED
PARTIALLY_RECEIVED
RECEIVED
CANCELLED

FR-PUR-004

A purchase order shall not itself increase physical stock.

14.2 Goods Receipt

FR-GRN-001

Authorized users shall be able to record goods received.

FR-GRN-002

A goods receipt may reference a purchase order.

FR-GRN-003

A purchase order shall not be mandatory unless business policy requires it.

FR-GRN-004

Posting a goods receipt shall create appropriate inventory movements.

FR-GRN-005

Receiving shall support partial quantities.

FR-GRN-006

Serial-tracked goods shall require serial capture.

FR-GRN-007

Batch-tracked goods shall require batch allocation.

FR-GRN-008

A single receipt line shall support multiple batch allocations where necessary.

FR-GRN-009

A single receipt line shall support multiple serial allocations where necessary.

FR-GRN-010

The quantity represented by allocations shall reconcile with the receipt-line quantity.

14.3 Purchase Invoice

FR-PINV-001

Authorized users shall be able to record purchase invoices.

FR-PINV-002

Purchase invoices shall contain:

- Supplier
- Invoice number
- Invoice date
- Items
- Quantity
- Unit price
- Discount
- Tax
- Total
- Status

FR-PINV-003

A purchase invoice may be related to one or more receipt records according to workflow.

FR-PINV-004

Purchase invoice posting shall affect supplier payable state.

FR-PINV-005

Purchase invoice creation shall not duplicate physical stock receipt when stock was already received through a goods receipt.

---

15. Inventory

Inventory is a core domain.

All stock-affecting operations shall use the central inventory engine.

FR-INV-001 — Central Inventory Engine

Sales, purchases, returns, transfers, adjustments, damage, and internal-use operations shall use controlled inventory logic.

FR-INV-002 — Movement Ledger

The system shall maintain an append-oriented inventory movement ledger.

FR-INV-003 — Balance Projection

The system shall maintain current stock balances as a transactional projection for fast reads.

FR-INV-004 — Stock Sources

Valid stock-in sources may include:

- Opening stock
- Purchase receipt
- Sales return
- Transfer in
- Approved adjustment
- Approved repair return where applicable

FR-INV-005 — Stock Uses

Valid stock-out sources may include:

- Retail sale
- Wholesale sale
- Dealer sale
- Purchase return
- Transfer out

- Damage
- Internal use
- Approved adjustment
- Repair movement where applicable

FR-INV-006 — Movement Traceability

Each movement shall identify its originating transaction/source.

FR-INV-007 — Warehouse/Location

Each stock position shall be associated with the applicable warehouse/location.

FR-INV-008 — Available Stock

The system shall expose stock availability according to quantity and reservation rules.

FR-INV-009 — Negative Stock

The default behavior shall prevent negative available stock.

Any exception must be explicitly authorized by confirmed business policy.

FR-INV-010 — No Direct Stock Editing

Users shall not directly overwrite the authoritative stock ledger.

FR-INV-011 — Adjustment

Stock adjustments shall create controlled adjustment movements and retain the reason.

FR-INV-012 — Historical Immutability

Historical inventory movements shall not be silently edited or deleted.

FR-INV-013 — Concurrency

Concurrent stock operations shall be protected so that the same available stock cannot be consumed twice.

FR-INV-014 — Transfer Integrity

Transfers shall not create or destroy net inventory.

---

16. Serial Number Management

FR-SERIAL-001

The system shall support registration of serialized physical units.

FR-SERIAL-002

Serial numbers shall be unique within the approved uniqueness scope.

FR-SERIAL-003

A serial-tracked receipt shall capture the required serial identity.

FR-SERIAL-004

A serial shall have a controlled lifecycle state.

Candidate states:

RECEIVED
IN_STOCK
RESERVED
SOLD
RETURNED
REPAIR
DAMAGED
SCRAPPED

Final lifecycle is **TBD — Business Confirmation Required**.

FR-SERIAL-005

Only an eligible serial may be sold.

FR-SERIAL-006

A serialized sale shall record the selected serial.

FR-SERIAL-007

Serial movement shall preserve physical-unit identity.

FR-SERIAL-008

The system shall retain serial movement history across:

- Receipt
- Transfer
- Sale
- Return
- Repair
- Damage
- Scrap

where applicable.

FR-SERIAL-009

Concurrent requests shall prevent the same serial from being sold/consumed twice.

---

17. Batch Management

FR-BATCH-001

The system shall support batch creation for batch-tracked products.

FR-BATCH-002

A batch shall support:

- Batch number
- Product variant
- Manufacturing date where applicable
- Expiry date where applicable

FR-BATCH-003

Batch identity shall be preserved through applicable inventory movements.

FR-BATCH-004

A document line may contain multiple batch allocations.

FR-BATCH-005

Batch allocation quantities shall reconcile with the parent document line.

FR-BATCH-006

Expiry-sensitive stock shall follow configured expiry rules.

Exact expiry policy is **TBD — Business Confirmation Required**.

---

18. Stock Transfers

FR-TRANSFER-001

Authorized users shall be able to create stock transfers.

FR-TRANSFER-002

Transfers shall identify:

- Source warehouse
- Destination warehouse
- Product
- Quantity
- Batch/serial identity where applicable

FR-TRANSFER-003

A transfer shall create corresponding stock movements.

FR-TRANSFER-004

Serialized transfers shall preserve serial identity.

FR-TRANSFER-005

Batch-tracked transfers shall preserve batch identity.

FR-TRANSFER-006

Source stock shall be validated before transfer posting.

FR-TRANSFER-007

The system shall prevent transfer to the same warehouse.

Whether transfers require an in-transit state is **TBD — Business Confirmation Required**.

---

19. Stock Adjustments

FR-ADJ-001

Authorized users shall be able to initiate stock adjustments.

FR-ADJ-002

Every adjustment shall require a reason.

FR-ADJ-003

Sensitive adjustments may require approval.

FR-ADJ-004

Posting an adjustment shall create inventory movements.

FR-ADJ-005

Adjustments shall not overwrite historical movement records.

---

20. Retail POS

FR-POS-001

Authorized sales users shall be able to open POS.

FR-POS-002

POS shall support fast product search.

FR-POS-003

POS shall support barcode scanning.

FR-POS-004

POS shall support customer selection or a configured walk-in customer.

FR-POS-005

POS shall support quantity entry.

FR-POS-006

POS shall support applicable serial selection.

FR-POS-007

POS shall support applicable batch selection.

FR-POS-008

POS shall apply the configured price.

FR-POS-009

POS shall support discounts.

FR-POS-010

POS shall calculate applicable taxes.

FR-POS-011

POS shall support applicable payment methods.

FR-POS-012

Completing a valid POS sale shall create the normal sales transaction.

FR-POS-013

Completing a valid stock sale shall create the appropriate inventory movement.

FR-POS-014

POS shall use the same backend sales engine as non-POS sales.

There shall not be a separate POS stock database.

FR-POS-015

A failed Tally synchronization shall not invalidate a successfully committed POS sale.

FR-POS-016

POS must provide clear success/failure feedback.

---

21. Wholesale and Dealer Sales

FR-SALE-001

Authorized users shall be able to create wholesale/dealer sales.

FR-SALE-002

Sales shall support party selection.

FR-SALE-003

Sales shall use the applicable price-list rules.

FR-SALE-004

Sales shall support credit transactions.

FR-SALE-005

Credit-limit rules shall be enforced.

Exact behavior is **TBD — Business Confirmation Required**.

Candidate behaviors:

BLOCK
WARN
REQUIRE AUTHORIZATION
ALLOW

FR-SALE-006

Unpaid/partially paid sales shall create the appropriate receivable.

FR-SALE-007

Retail, wholesale, and dealer sales shall share the same core sales engine.

---

22. Quotations

FR-QUOTE-001

Authorized users shall be able to create quotations.

FR-QUOTE-002

Quotations shall support:

- Party
- Items
- Quantities
- Prices
- Discounts
- Taxes
- Validity
- Status
- Notes

FR-QUOTE-003

Quotation creation shall not reduce physical stock by default.

FR-QUOTE-004

Quotation-to-order conversion shall preserve traceability.

Quotation locking/version behavior is **TBD — Business Confirmation Required**.

---

23. Sales Orders

FR-ORDER-001

Authorized users shall be able to create sales orders.

FR-ORDER-002

Sales orders shall support controlled lifecycle states.

Candidate states:

DRAFT
CONFIRMED
PARTIALLY_FULFILLED
FULFILLED
CANCELLED

FR-ORDER-003

A sales order may be converted into one or more invoices according to approved workflow.

FR-ORDER-004

Partial fulfilment/invoicing shall be supported where required.

FR-ORDER-005

The system shall preserve line-level traceability from order to invoice.

FR-ORDER-006

Stock reservation behavior shall be configurable according to approved policy.

Reservation policy is **TBD — Business Confirmation Required**.

---

24. Sales Invoices

FR-INVOICE-001

The system shall create sales invoices from supported sales workflows.

FR-INVOICE-002

Invoices shall contain:

- Invoice number
- Date
- Party
- Items
- Quantity
- Unit price
- Discount
- Tax
- Total
- Payment status

FR-INVOICE-003

Invoice numbers shall be generated using concurrency-safe numbering.

FR-INVOICE-004

Posted invoices shall not be silently edited or deleted.

FR-INVOICE-005

Authorized users shall be able to cancel/reverse invoices according to business rules.

FR-INVOICE-006

Cancellation/reversal shall preserve the original invoice and audit trail.

FR-INVOICE-007

Invoice totals shall be deterministic and testable.

FR-INVOICE-008

Historical prices shall remain unchanged when current price lists change.

---

25. Payments

FR-PAY-001

Payments shall be recorded independently of invoices.

FR-PAY-002

The system shall support applicable payment methods such as:

- Cash
- UPI/digital payment
- Bank transfer
- Card
- Cheque
- Approved credit/adjustment

FR-PAY-003

The system shall support partial payments.

FR-PAY-004

The system shall support advances where required.

FR-PAY-005

A payment may be allocated to multiple eligible invoices.

FR-PAY-006

An invoice may be settled by multiple payments.

FR-PAY-007

Payment allocations shall be validated against eligible balances.

FR-PAY-008

Payment history shall remain traceable.

FR-PAY-009

Payment reversal shall be controlled and audited.

Overpayment and advance application rules are **TBD — Business Confirmation Required**.

---

26. Receivables and Payables

FR-AR-001

The system shall show customer/dealer outstanding balances.

FR-AR-002

The system shall show supplier payable balances.

FR-AR-003

The system shall support due-date information where payment terms exist.

FR-AR-004

The system shall identify overdue balances where configured.

FR-AR-005

Outstanding balances shall be derived from authoritative posted transactions and allocations.

FR-AR-006

Users shall be able to inspect the transactions contributing to an outstanding balance.

---

27. Sales Returns

FR-SRET-001

Authorized users shall be able to create sales returns.

FR-SRET-002

A sales return should reference the original sales invoice where applicable.

FR-SRET-003

Returned quantity shall be validated against eligible original quantity.

FR-SRET-004

Serialized returns shall identify the returned serial unit.

FR-SRET-005

Batch-tracked returns shall identify the applicable batch.

FR-SRET-006

Returned items shall have a disposition.

Candidate dispositions:

RESELLABLE
REPAIR_REQUIRED
DAMAGED
SCRAP
SUPPLIER_RETURN

FR-SRET-007

Inventory effect shall depend on the approved disposition.

FR-SRET-008

Financial effect shall be represented by an appropriate return/credit transaction.

FR-SRET-009

Return history shall remain linked to the original sale.

Return inspection rules are **TBD — Business Confirmation Required**.

---

28. Purchase Returns

FR-PRET-001

Authorized users shall be able to create purchase returns.

FR-PRET-002

A purchase return should reference the original purchase where applicable.

FR-PRET-003

Returned quantity shall be validated against eligible received/purchased quantity.

FR-PRET-004

Batch/serial identity shall be preserved where applicable.

FR-PRET-005

Posting a purchase return shall reduce the appropriate inventory.

FR-PRET-006

Supplier payable/financial impact shall be recorded.

---

29. Delivery and Fulfilment

FR-DEL-001

A sale/order may require delivery.

FR-DEL-002

A delivery shall support:

- Customer/party
- Source order/invoice
- Delivery address
- Assigned staff
- Scheduled time
- Status
- Delivery time
- Notes
- Delivery items

FR-DEL-003

Candidate statuses:

PENDING
READY
OUT_FOR_DELIVERY
DELIVERED
CANCELLED

FR-DEL-004

The system shall support multiple delivery items.

FR-DEL-005

Partial fulfilment shall be supported where required.

FR-DEL-006

Serialized delivery items shall preserve serial identity where applicable.

FR-DEL-007

Delivery completion shall record delivery time and relevant notes.

The exact point at which stock is deducted versus dispatched/delivered is **TBD — Business Confirmation Required**.

---

30. Installation

Installation is optional in the initial implementation.

FR-INST-001

A sale/delivery may indicate that installation is required.

FR-INST-002

The system shall support installation scheduling.

FR-INST-003

Candidate statuses:

PENDING
SCHEDULED
IN_PROGRESS
COMPLETED
CANCELLED

FR-INST-004

Installation shall reference the customer and relevant product/sale.

FR-INST-005

Installation shall support multiple items where required.

Advanced technician/checklist/service workflow is outside MVP unless approved.

---

31. Warranty

FR-WAR-001

The system shall support warranty records for applicable products.

FR-WAR-002

Warranty may be associated with:

- Product variant
- Serial unit
- Sales invoice
- Party

FR-WAR-003

The system shall store warranty start and expiry dates.

FR-WAR-004

Serialized products should normally have warranty traceability through the serial unit.

FR-WAR-005

The system shall support warranty status.

Warranty start-date policy is **TBD — Business Confirmation Required**.

---

32. Document Lifecycle

Important transaction documents shall have controlled lifecycle states.

Generic model:

DRAFT
↓
CONFIRMED
↓
POSTED
↓

COMPLETED

Correction path:

POSTED
  ├── CANCELLED
  ├── REVERSED
  └── CORRECTED THROUGH NEW TRANSACTION

FR-LIFE-001

The system shall prevent invalid state transitions.

FR-LIFE-002

Posted documents shall not be silently deleted.

FR-LIFE-003

Cancellation and reversal shall require appropriate permission.

FR-LIFE-004

Inventory/financial effects of cancellation shall be handled through controlled compensating transactions.

FR-LIFE-005

Lifecycle changes shall be auditable.

Exact lifecycle states are defined per module during detailed design.

---

33. Document Numbering

FR-NUM-001

Each numbered document type shall have a concurrency-safe numbering mechanism.

FR-NUM-002

Document numbers shall be unique within the approved organization/branch/document-type scope.

FR-NUM-003

Document numbering shall not depend on frontend-generated counters.

FR-NUM-004

Numbering configuration shall support future fiscal-year rules.

Exact:

- prefix
- fiscal-year reset
- branch scope
- gap policy
- manual numbering

are **TBD — Business Confirmation Required**.

---

34. Audit Logging

FR-AUDIT-001

Critical business actions shall generate audit records.

FR-AUDIT-002

Audit records shall capture, where applicable:

- User
- Action
- Entity
- Entity ID
- Timestamp
- Organization/branch
- Before value
- After value
- Reason
- Correlation ID
- Related transaction

FR-AUDIT-003

Audit records shall be protected from ordinary user modification.

FR-AUDIT-004

Security-sensitive actions shall be audited.

FR-AUDIT-005

Audit logs shall not store secrets or unnecessary sensitive payloads.

---

35. Search and Filtering

The system shall provide consistent search/filter behavior across major modules.

Supported search dimensions may include:

Product name
SKU
Barcode
Model
Serial number
Batch number
Party name
Invoice number
Order number
PO number
Date range
Status
Warehouse
Branch
Brand
Category

FR-SEARCH-001

Search shall support pagination for large result sets.

FR-SEARCH-002

Exact identifiers such as SKU/barcode/serial number shall support deterministic matching.

FR-SEARCH-003

Search results shall respect authorization scope.

FR-SEARCH-004

POS product search shall be optimized for frequent use.

---

36. Attachments and Documents

FR-FILE-001

The system shall support attachments for appropriate business records.

Possible records:

- Purchase documents
- Sales documents
- Party documents
- Warranty documents
- Delivery documents
- Other approved records

FR-FILE-002

Attachment access shall follow business authorization.

FR-FILE-003

Actual file binaries should normally be stored in object storage.

FR-FILE-004

Uploaded files shall be validated according to security policy.

---

37. Notifications

FR-NOT-001

The system shall provide in-app notifications.

Initial notification candidates:

- Low stock
- Overdue payment
- Credit limit warning
- Pending approval
- Pending delivery
- Warranty expiry
- Failed Tally synchronization

FR-NOT-002

Notifications shall not become the source of truth for business state.

FR-NOT-003

Failed background operations shall remain visible through operational status screens even if a notification is missed.

---

38. Reports

Reports must use authoritative ERP data and respect authorization.

FR-REP-001 — Sales

The system shall support:

- Daily sales
- Monthly sales
- Product-wise sales
- Category-wise sales
- Brand-wise sales
- Party-wise sales
- Retail vs wholesale/dealer sales
- Sales returns
- Discount analysis where configured

FR-REP-002 — Inventory

The system shall support:

- Current stock
- Stock movement
- Location-wise stock
- Low stock
- Stock valuation
- Stock ageing
- Fast-moving products
- Slow-moving products
- Batch/serial stock where applicable

FR-REP-003 — Parties

The system shall support:

- Customer ledger
- Dealer ledger
- Supplier ledger
- Receivables
- Payables
- Payment history

FR-REP-004 — Procurement

The system shall support:

- Purchase history
- Supplier-wise purchases
- Product-wise purchases
- Purchase returns
- Receipt history

FR-REP-005 — Operational Exceptions

The system should provide visibility into:

- Failed Tally sync
- Pending approvals
- Negative-stock attempts
- Unallocated/exceptional inventory
- Overdue receivables
- Other important operational exceptions

FR-REP-006 — Dashboard

The dashboard shall display KPIs appropriate to the user's permissions.

Final report catalog is **TBD — Business Confirmation Required**.

---

39. TallyPrime Integration

Tally is an external accounting boundary.

FR-TALLY-001 — Configuration

Authorized administrators shall be able to configure Tally integration.

FR-TALLY-002 — Master Mapping

The system shall maintain ERP-to-Tally mappings.

Examples:

ERP Party → Tally Ledger
ERP Product → Tally Stock Item
ERP Tax/Accounting Configuration → Tally Configuration

FR-TALLY-003 — Sales Sync

Eligible sales shall generate durable synchronization work.

FR-TALLY-004 — Purchase Sync

Eligible purchases shall generate durable synchronization work.

FR-TALLY-005 — Receipt Sync

Eligible customer/dealer receipts shall generate synchronization work.

FR-TALLY-006 — Payment Sync

Eligible supplier/business payments shall generate synchronization work.

FR-TALLY-007 — Asynchronous Processing

Tally synchronization shall not block the core ERP transaction.

FR-TALLY-008 — Durable Event

A successful core transaction that requires integration shall create a durable integration event as part of the same database transaction.

FR-TALLY-009 — Retry

Failed synchronization shall be retryable.

FR-TALLY-010 — Idempotency

The same ERP event shall not create duplicate Tally transactions.

FR-TALLY-011 — Sync Status

Synchronization shall expose statuses such as:

PENDING
PROCESSING
SYNCED
FAILED
RETRYING

FR-TALLY-012 — Error Visibility

Useful error information shall be available to authorized users.

FR-TALLY-013 — Reconciliation

Authorized users shall be able to identify ERP transactions whose Tally representation is missing, failed, or inconsistent.

FR-TALLY-014 — ERP Independence

Tally unavailability shall not prevent a valid ERP operational transaction from committing.

Exact Tally communication method, mappings, voucher behavior, and synchronization direction are **TBD — Business Confirmation Required** and belong in `TALLY-INTEGRATION.md`.

---

40. Data Integrity Requirements

NFR-DATA-001 — Atomicity

Business transactions affecting multiple records shall commit atomically.

NFR-DATA-002 — Rollback

A failed transaction shall not leave partial inventory or financial state.

NFR-DATA-003 — Referential Integrity

Core relationships shall use database foreign keys where practical.

NFR-DATA-004 — Historical Integrity

Posted transactions shall remain traceable.

NFR-DATA-005 — Derived Data

Derived balances shall be rebuildable from authoritative data where practical.

NFR-DATA-006 — Money Precision

Money shall use exact decimal/numeric representation.

NFR-DATA-007 — Quantity Precision

Quantities shall support the precision required by their unit.

NFR-DATA-008 — Organization Isolation

Cross-organization data relationships shall be rejected.

NFR-DATA-009 — Concurrency

Critical operations shall be safe under concurrent requests.

---

41. Security Requirements

NFR-SEC-001

Passwords shall be securely hashed using the approved password hashing mechanism.

NFR-SEC-002

Production communication shall use HTTPS.

NFR-SEC-003

Authorization shall be enforced server-side.

NFR-SEC-004

Object-level authorization shall be enforced for scoped business records.

NFR-SEC-005

Secrets shall not be stored in source code.

NFR-SEC-006

Sensitive configuration shall use approved secret-management mechanisms.

NFR-SEC-007

Security-sensitive actions shall be audited.

NFR-SEC-008

Database access shall use least privilege.

NFR-SEC-009

File uploads shall be validated and protected against unauthorized access.

NFR-SEC-010

Tally credentials shall never be stored in plaintext.

NFR-SEC-011

The application shall not trust client-supplied organization/branch/warehouse scope without server verification.

Detailed controls are governed by `PROTECTION.md`.

---

42. Performance Requirements

The system must be responsive under expected ERP/POS workload.

NFR-PERF-001

Common CRUD/navigation operations should normally respond within an acceptable interactive range under expected load.

NFR-PERF-002

POS search shall be optimized for frequent queries.

NFR-PERF-003

Large report queries shall support filtering and pagination.

NFR-PERF-004

Long-running jobs shall not block normal user interaction.

NFR-PERF-005

Inventory reads shall use the balance projection rather than repeatedly recalculating large ledgers when the projection is available.

NFR-PERF-006

Indexes shall be based on actual query patterns.

No arbitrary millisecond SLA is defined yet.

Performance targets shall be finalized after production-like workload testing.

---

43. Availability and Reliability

NFR-REL-001

The system shall minimize committed transaction loss.

NFR-REL-002

Background integration work shall survive process restarts.

NFR-REL-003

Failed integration work shall support retry.

NFR-REL-004

Critical failures shall be logged with correlation information.

NFR-REL-005

Database backups shall exist before production launch.

NFR-REL-006

Restore procedures shall be documented and tested.

NFR-REL-007

ERP operation shall remain usable when external services such as Tally are temporarily unavailable.

---

44. Scalability

NFR-SCALE-001

The system shall support additional users without redesign of core modules.

NFR-SCALE-002

The system shall support additional products and transaction volume.

NFR-SCALE-003

The system shall support additional warehouses and branches.

NFR-SCALE-004

The modular monolith shall preserve clear module boundaries.

NFR-SCALE-005

The architecture shall allow future extraction of services only when an actual scaling/ownership/reliability need is demonstrated.

---

45. Usability

NFR-UX-001

The ERP shall be understandable to non-technical business users.

NFR-UX-002

Common operations shall avoid unnecessary steps.

NFR-UX-003

POS shall prioritize speed and low interaction overhead.

NFR-UX-004

Forms shall show clear validation messages.

NFR-UX-005

The system shall provide useful:

- Loading states
- Empty states
- Success states
- Error states
- Confirmation states

NFR-UX-006

Destructive/sensitive operations shall require clear confirmation where appropriate.

---

46. Accessibility and Responsive Behavior

NFR-UI-001

Core ERP screens shall support common desktop screen sizes.

NFR-UI-002

Important operational screens should remain usable on smaller screens where practical.

NFR-UI-003

Interactive controls shall have clear labels and states.

NFR-UI-004

Keyboard-efficient workflows should be supported in POS where practical.

---

47. Observability

NFR-OBS-001

Important requests shall have correlation identifiers.

NFR-OBS-002

Critical workflows shall be traceable across:

Request
↓
Application Use Case
↓
Database Transaction
↓
Audit
↓
Integration Event
↓
Tally Sync Job
↓
Sync Attempt

NFR-OBS-003

Logs shall not expose secrets.

NFR-OBS-004

Operational failures shall provide actionable diagnostics.

---

48. Backup and Recovery

Production shall implement:

- Scheduled database backups
- Backup retention
- Backup verification
- Recovery procedure
- Restore testing
- Migration recovery procedure
- Disaster recovery documentation

Target:

RPO: TBD
RTO: TBD

These values must be chosen from actual business requirements rather than invented.

---

49. API and Integration Behavior

The API shall:

1. authenticate requests;
2. authorize operations;
3. validate input;
4. enforce business rules;
5. enforce organization/branch/location scope;
6. execute appropriate use cases;
7. perform required database transactions;
8. create audit records;
9. create integration events where required;
10. return deterministic errors.

Imports and background jobs shall use the same business/domain rules as normal API requests.

No import path or internal job may bypass inventory, financial, authorization, or audit controls.

---

50. Error Handling

The system shall return clear, safe errors.

Errors should distinguish between:

Validation error
Authorization error
Not found
Conflict/concurrency failure
Business rule violation
Integration failure
Unexpected system failure

Sensitive implementation details must not be exposed to end users.

Errors shall be correlated for support/debugging.

---

51. Idempotency

Critical mutation operations should support idempotency where duplicate requests are reasonably possible.

Examples:

- payment creation
- invoice posting
- inventory posting
- integration events
- Tally synchronization
- import jobs

A repeated request with the same valid idempotency key shall not create unintended duplicate business effects.

---

52. Import and Export

FR-IMP-001

Authorized users may import approved master/transaction data.

FR-IMP-002

Imports shall validate data before committing.

FR-IMP-003

Imports shall not bypass business rules.

FR-IMP-004

Large imports shall provide progress/error reporting.

FR-IMP-005

Import failures shall not leave partially committed business transactions where atomicity is required.

FR-EXP-001

Authorized users may export approved reports/data.

FR-EXP-002

Exports shall respect authorization scope.

FR-EXP-003

Sensitive data exports shall be permission-controlled and auditable where required.

---

53. Transaction Acceptance Criteria

Every important feature must have acceptance criteria.

53.1 Retail Sale

A retail sale is accepted when:

1. An authorized user can create it.
2. Product search/scan works.
3. Required serial/batch information can be selected.
4. Correct pricing is applied.
5. Applicable tax is calculated.
6. Stock availability is validated.
7. Invoice is generated.
8. Payment is recorded where applicable.
9. Inventory is reduced exactly once.
10. Audit information exists.
11. Integration event exists when required.
12. Tally failure does not roll back the ERP sale.

53.2 Purchase Receipt

Accepted when:

1. Authorized user can record receipt.
2. Supplier and warehouse are valid.
3. PO linkage works when supplied.
4. Partial receipt works.
5. Serial/batch allocations are captured where required.

6. Inventory increases exactly once.
7. Balance projection updates correctly.
8. Audit trail exists.
9. Integration event is created where applicable.

53.3 Stock Transfer

Accepted when:

1. Source stock is validated.
2. Destination is valid.
3. Batch/serial identity is preserved.
4. Source quantity decreases correctly.
5. Destination quantity increases correctly.
6. Net stock is unchanged.
7. Concurrent transfers cannot consume the same stock twice.
8. Transfer history is traceable.

53.4 Payment

Accepted when:

1. Authorized user can create payment.
2. Payment method is recorded.
3. Partial payment works.
4. Allocation to invoice works.
5. Multiple allocations work.
6. Outstanding recalculates correctly.
7. Duplicate allocation is prevented.
8. Audit history exists.
9. Tally event is created when applicable.

---

54. Non-Functional Acceptance

The product is not production-ready merely because screens work.

A release must also demonstrate:

- Correct authorization
- Correct database constraints
- Transaction atomicity
- Inventory concurrency protection
- Historical integrity
- Auditability

- Backup/recovery readiness
- Integration retry behavior
- Idempotency
- Error handling
- Performance under expected workload
- Security testing appropriate to release risk

---

55. MVP Priority

P0 — Mandatory

- Authentication
- Users/roles/permissions
- Organization/branch/warehouse
- Products
- Parties
- Pricing
- Purchases
- Goods receipts
- Inventory
- Serial/batch foundation
- Retail POS
- Wholesale/dealer sales
- Sales invoices
- Payments
- Receivables/payables
- Returns
- Core reports
- Audit
- Tally synchronization foundation

P1 — Important

- Quotations
- Sales orders
- Delivery
- Installation
- Warranty
- Advanced inventory reports
- Advanced reconciliation
- Notifications
- Attachments

P2 — Future

- WhatsApp
- Advanced CRM
- Service management
- AI analytics
- Native mobile application
- E-commerce
- Marketplace integrations
- Commercial SaaS features

---

56. Requirements Traceability

Every P0/P1 requirement should eventually map to:

BRD
↓
SRS Requirement ID
↓
Business Rule
↓
Use Case
↓
API
↓
Database Entity
↓
UI
↓
Automated Test
↓
Acceptance Test

Example:

FR-INV-013
↓
BR-CORE / BR-INVENTORY rules
↓
Sell Inventory Use Case
↓
POST /sales
↓
sales_invoices + inventory_movements + inventory_balances

↓
POS Sales Screen
↓
Concurrency Test
↓
Acceptance: one unit cannot be sold twice

---

57. Requirement IDs and Change Control

Requirement IDs are stable identifiers.

When a requirement changes:

1. retain the requirement ID where the underlying requirement remains the same;
2. document the change;
3. record reason;
4. record impact;
5. update affected design documents;
6. update tests;
7. obtain required approval.

Example change record:

Requirement: FR-INV-010
Change: Negative stock changed from blocked to manager-authorized exception
Reason: Approved business policy
Impact: Inventory service + permissions + audit + tests
Approval: TBD
Date: TBD

---

58. Open Business Decisions

The following must be confirmed before final implementation freeze:

1. Exact operational workflows.
2. Branch/warehouse structure.
3. GST/tax structure.
4. Invoice format.
5. Document numbering.
6. Fiscal-year numbering behavior.
7. Negative-stock exception policy.
8. Stock valuation method.
9. Serial uniqueness scope.

10. Serial lifecycle states.
11. Batch/expiry policy.
12. FEFO/FIFO behavior if required.
13. Sales-order reservation policy.
14. Credit-limit behavior.
15. Payment overpayment behavior.
16. Advance-payment behavior.
17. Warranty start date.
18. Return inspection process.
19. Delivery stock-deduction timing.
20. Transfer in-transit behavior.
21. Installation workflow.
22. Approval thresholds.
23. Branch/location access rules.
24. Purchase-order mandatory/optional policy.
25. Multiple receipt/invoice matching.
26. Partial sales-order invoicing.
27. Quotation locking/versioning.
28. Cancellation after payment.
29. Cancellation after delivery.
30. Cancellation after Tally synchronization.
31. Damaged/internal-use workflow.
32. Receivable/payable calculation rules.
33. Rounding rules.
34. Currency requirements.
35. Tally version/configuration.
36. Tally communication method.
37. ERP-to-Tally master mapping.
38. ERP-to-Tally voucher mapping.
39. Tally synchronization direction.
40. Tally reconciliation/correction policy.
41. Data retention.
42. Backup RPO.
43. Backup RTO.
44. Final report catalog.

These are decisions, not implementation details. They must not be silently invented by developers.

---

59. Definition of Done

A feature/module is complete only when:

- SRS requirements are implemented;
- relevant business rules are enforced;

- backend validation exists;
- authorization exists;
- database constraints are appropriate;
- transaction boundaries are correct;
- concurrency behavior is tested;
- errors are handled;
- audit requirements are satisfied;
- relevant reports are correct;
- integrations are tested where applicable;
- automated tests pass;
- documentation is updated;
- acceptance criteria pass.

A working UI alone is not Definition of Done.

---

60. Cross-Document Responsibilities

Document	Responsibility
`BRD.md`	Business needs/objectives
`SRS.md`	Software requirements/behavior
`BUSINESS-RULES_v1.0.md`	Business policy/invariants
`ARCHITECTURE_v2.0.md`	System structure/technical architecture
`DATABASE_v1.0.md`	Data model/database architecture
`TECHSTACK_v1.1.md`	Technology choices
`API.md`	API contracts
`TALLY-INTEGRATION.md`	Tally-specific integration
`PROTECTION.md`	Detailed security/protection
`DESIGN.md`	UI/UX
`PHASES.md`	Delivery roadmap
`TESTING.md`	Testing strategy

If documents conflict, the conflict must be explicitly resolved rather than silently choosing an interpretation.

---

61. Implementation Sequence

Recommended sequence:

SRS
↓
Business Rules Confirmation
↓
Use Cases / User Stories
↓
Architecture
↓
Database
↓
ERD
↓
API Specification
↓
UI/UX
↓
Implementation
↓
Automated Tests
↓
Staging
↓
Production

For implementation:

Foundation
↓
Auth / Authorization
↓
Masters
↓
Inventory Engine
↓
Purchasing
↓
Sales / POS
↓

Payments
↓
Returns
↓
Fulfilment
↓
Audit / Reporting
↓
Tally

The inventory engine must be treated as a core foundation, not as a later add-on.

---

62. Final Non-Negotiable Requirements

1. PostgreSQL is the ERP transactional source of truth.
2. Business rules are server-enforced.
3. Authorization is server-enforced.
4. Inventory uses a controlled movement ledger.
5. Inventory balances are projections, not free-edit fields.
6. Serial identity cannot be duplicated or consumed twice.
7. Batch identity must be preserved where required.
8. Posted transactions are historical facts.
9. Corrections use controlled cancellation/reversal/return/adjustment flows.
10. Financial allocations use strong integrity.
11. Critical transactions are atomic.
12. Concurrent stock operations are protected.
13. Tally cannot block a valid ERP transaction.
14. Tally integration is durable, retryable, and idempotent.
15. Audit records are protected.
16. Organization/branch/location scope is enforced.
17. Money uses exact decimal/numeric values.
18. Imports cannot bypass business rules.
19. The ERP remains generic and configurable.
20. Infrastructure complexity must be justified by actual requirements.